

# Changement de variable

$$\frac{A \partial^2 u}{\partial^2 x} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial^2 y} = 0$$

$$\varphi = \alpha y + \beta x \quad \Leftrightarrow \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\theta = \gamma y + \delta x$$

$$\alpha \delta - \gamma \beta \neq 0 \quad \text{invertible.}$$

$$u(x, y) = u(x(\varphi, \theta), y(\varphi, \theta))$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial x} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial x}$$

$$\frac{\partial u}{\partial x} = \beta \frac{\partial u}{\partial \varphi} + \delta \frac{\partial u}{\partial \theta}$$

$$\boxed{\frac{\partial^2 u}{\partial^2 x}} = \beta \frac{\partial}{\partial x} \frac{\partial u}{\partial \varphi} + \delta \frac{\partial^2 u}{\partial^2 \theta}$$

$$= \beta \frac{\partial}{\partial \varphi} \left( \frac{\partial u}{\partial x} \right) + \delta \frac{\partial}{\partial \theta} \left( \frac{\partial u}{\partial x} \right)$$

$$= \beta \frac{\partial}{\partial \varphi} \left( \beta \frac{\partial u}{\partial \varphi} + \delta \frac{\partial u}{\partial \theta} \right) + \delta \frac{\partial}{\partial \theta} \left( \beta \frac{\partial u}{\partial \varphi} + \delta \frac{\partial u}{\partial \theta} \right)$$

$$= \beta^2 \frac{\partial^2 u}{\partial^2 \varphi} + \beta \delta \left( \frac{\partial^2 u}{\partial \varphi \partial \theta} \right) + \delta \beta \frac{\partial^2 u}{\partial \theta \partial \varphi} + \delta^2 \left( \frac{\partial^2 u}{\partial^2 \theta} \right)$$

$$\boxed{= \beta^2 \left( \frac{\partial^2 u}{\partial^2 \varphi} \right) + 2\beta \delta \left( \frac{\partial^2 u}{\partial \varphi \partial \theta} \right) + \delta^2 \frac{\partial^2 u}{\partial^2 \theta}}$$

x A

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial \varphi} \frac{\partial \varphi}{\partial y} + \frac{\partial u}{\partial \theta} \frac{\partial \theta}{\partial y}$$

$$\frac{\partial u}{\partial y} = \alpha \frac{\partial u}{\partial \varphi} + \gamma \frac{\partial u}{\partial \theta}$$

$$\boxed{\frac{\partial^2 u}{\partial x \partial y}} = \frac{\partial}{\partial x} \left( \alpha \frac{\partial u}{\partial \varphi} + \gamma \frac{\partial u}{\partial \theta} \right)$$

$$= \alpha \cdot \frac{\partial}{\partial \varphi} \left( \frac{\partial u}{\partial x} \right) + \gamma \frac{\partial}{\partial \theta} \left( \frac{\partial u}{\partial x} \right)$$

$$= \alpha \frac{\partial}{\partial \varphi} \left( \beta \frac{\partial u}{\partial \varphi} + \delta \frac{\partial u}{\partial \theta} \right) + \gamma \frac{\partial}{\partial \theta} \left( \beta \frac{\partial u}{\partial \varphi} + \delta \frac{\partial u}{\partial \theta} \right)$$

$$\boxed{= \alpha \beta \frac{\partial^2 u}{\partial^2 \varphi} + (\alpha \delta + \gamma \beta) \frac{\partial^2 u}{\partial \varphi \partial \theta} + \gamma \delta \frac{\partial^2 u}{\partial^2 \theta}}$$

x B

$$\frac{\partial^2 u}{\partial^2 y} = \frac{\partial}{\partial y} \left( \alpha \frac{\partial u}{\partial \varphi} + \gamma \frac{\partial u}{\partial \theta} \right)$$

$$= \alpha \frac{\partial}{\partial \varphi} \left( \frac{\partial u}{\partial y} \right) + \gamma \frac{\partial}{\partial \theta} \left( \frac{\partial u}{\partial y} \right)$$

$$= \alpha \frac{\partial}{\partial \varphi} \left( \alpha \frac{\partial u}{\partial \varphi} + \gamma \frac{\partial u}{\partial \theta} \right) + \gamma \frac{\partial}{\partial \theta} \left( \alpha \frac{\partial u}{\partial \varphi} + \gamma \frac{\partial u}{\partial \theta} \right)$$

$$\boxed{= \alpha^2 \frac{\partial^2 u}{\partial^2 \varphi} + 2\alpha \gamma \frac{\partial^2 u}{\partial \varphi \partial \theta} + \gamma^2 \frac{\partial^2 u}{\partial^2 \theta}}$$

x C

$$F\left(\frac{\partial^2 u}{\partial^2 x}, \frac{\partial^2 u}{\partial x \partial y}, \frac{\partial^2 u}{\partial^2 y}, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}\right) = 0$$

u. courbe .

$$\frac{\partial u}{\partial x}$$

$$\frac{\partial u}{\partial y}$$

Conditions

Aux

Limites

$$\frac{\partial^2 u}{\partial^2 \varphi} \rightarrow A\beta^2 + B\alpha\beta + C\alpha^2 \quad (1)$$

$$\frac{\partial^2 u}{\partial \varphi \partial \theta} \rightarrow 2A\beta\delta + B(\alpha\delta + \delta\beta) + C(2\alpha\gamma) \quad (3)$$

$$\frac{\partial^2 u}{\partial^2 \theta} \rightarrow A\delta^2 + B\delta\gamma + C\gamma^2 \quad (2)$$

$$(1) \Rightarrow A\left(\frac{\beta}{\alpha}\right)^2 + B\left(\frac{\beta}{\alpha}\right) + C$$

$$(2) \Rightarrow A\left(\frac{\delta}{\gamma}\right)^2 + B\left(\frac{\delta}{\gamma}\right) + C$$

$$\Delta = B^2 - 4AC > 0$$

$$\frac{\beta}{\alpha} = \frac{-B + \sqrt{\Delta}}{2A}$$

$$\frac{\delta}{\gamma} = \frac{-B - \sqrt{\Delta}}{2A}$$

$$\alpha = \gamma = 2A$$

$$\beta = -B + \sqrt{\Delta}$$

$$\delta = -B - \sqrt{\Delta}$$

$$\alpha\delta - \gamma\beta \neq 0 \Rightarrow \delta \neq \beta$$

Hyperbolique

$$\boxed{B^2 - 4AC = 0} \text{ Alors } B^2 = 4AC$$

$$\frac{\beta}{\alpha} = \frac{-B}{2A} \Rightarrow \alpha = 2A$$

$$\beta = -B$$

$$(1) AB^2 - 2AB^2 + 4A^2C$$

$$A(B^2 - 4AC) = 0$$

$$(2) -2AB + B(2A) + 0 = 0$$

$$(3) -2AB\delta + B(2A\delta - \gamma\beta) + 2C(4A\gamma)$$

$$= -\gamma B^2 + 4\gamma AC$$

$$\times \boxed{B^2 - 4AC = 0} \left\{ \begin{array}{l} \frac{\partial^2 u}{\partial^2 \varphi} \text{ coeff} = 0 \\ \frac{\partial^2 u}{\partial \varphi \partial \theta} \text{ coeff} = 0 \end{array} \right.$$

parabolique

Elliptique

$$\times \boxed{B^2 - 4AC < 0} \quad B^2 - 4AC = \Delta$$

$$\alpha = \frac{2A}{\sqrt{-\Delta}} \quad \beta = \frac{-B}{\sqrt{-\Delta}}$$

$$\gamma = 0 \quad \delta = 1$$

$$(1): \frac{AB^2}{(-\Delta)} + \frac{B(-2AB)}{(-\Delta)} + \frac{C4A^2}{(-\Delta)}$$

$$= \frac{A}{-\Delta} (-B^2 + 4AC) = A$$

$$(2) \quad 2A\left(\frac{-B}{\sqrt{-\Delta}}\right) + B\left(\frac{2A}{\sqrt{-\Delta}}\right) = 0$$

(3) A

$$A\left(\frac{\partial^2 u}{\partial^2 \varphi} + \frac{\partial^2 u}{\partial^2 \theta}\right) = 0$$

$$\Delta u = 0 \quad (\text{laplacien})$$